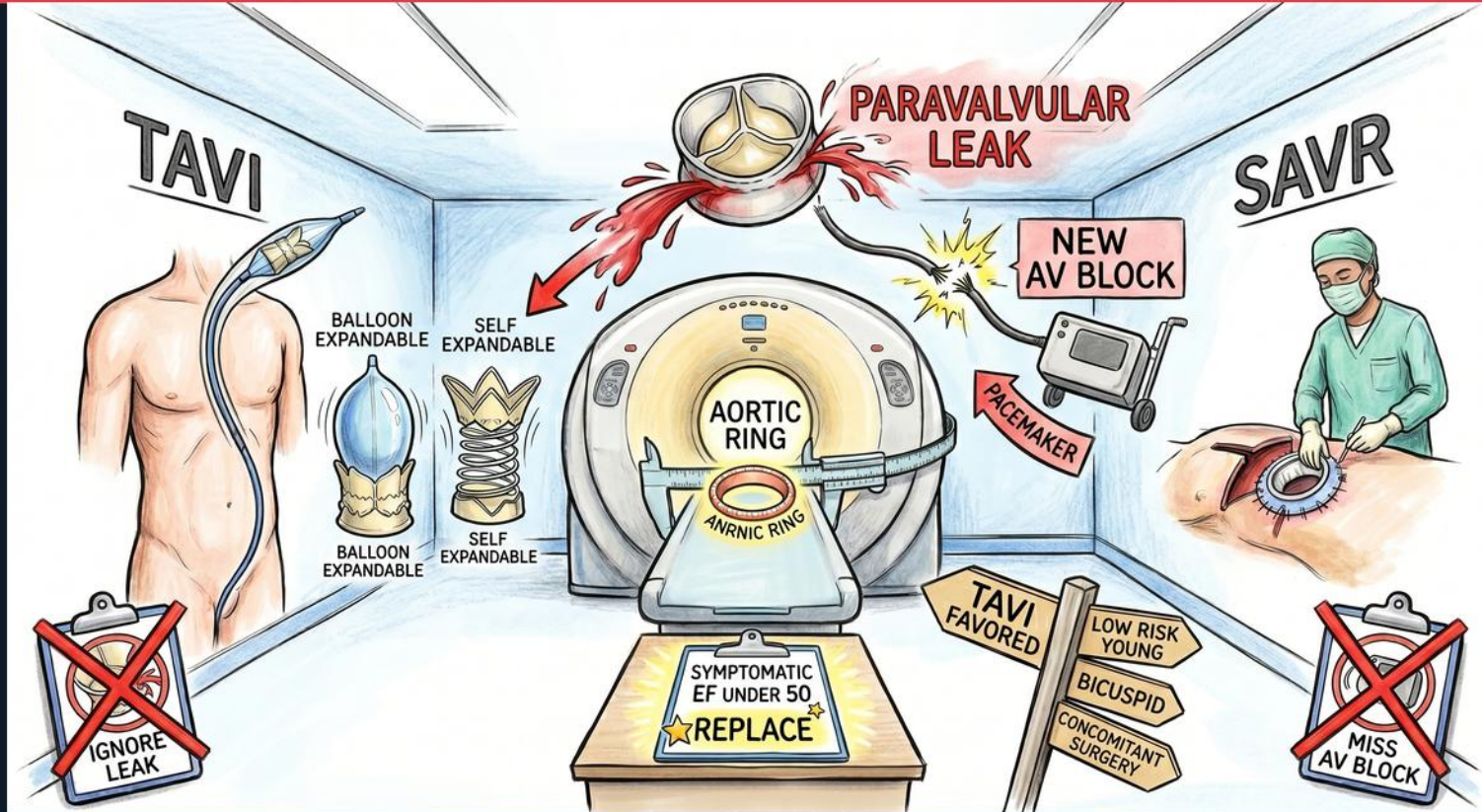


Valvular — TAVI vs SAVR





TAVI — transcatheter valve

WHAT YOU SEE

Left side labeled TAVI, a catheter threaded up the leg into the body — no chest incision.

WHAT IT ENCODES

Transcatheter aortic valve implantation delivers a prosthetic valve via a catheter (usually transfemoral), with no sternotomy and no cardiopulmonary bypass. The new valve is deployed inside the diseased native valve, so recovery is faster and it suits higher surgical-risk patients.

THE BOARD TRAP

Describing TAVI as open surgery — it is catheter-based valve-in-native deployment, distinct from SAVR.



SAVR — surgical replacement

WHAT YOU SEE

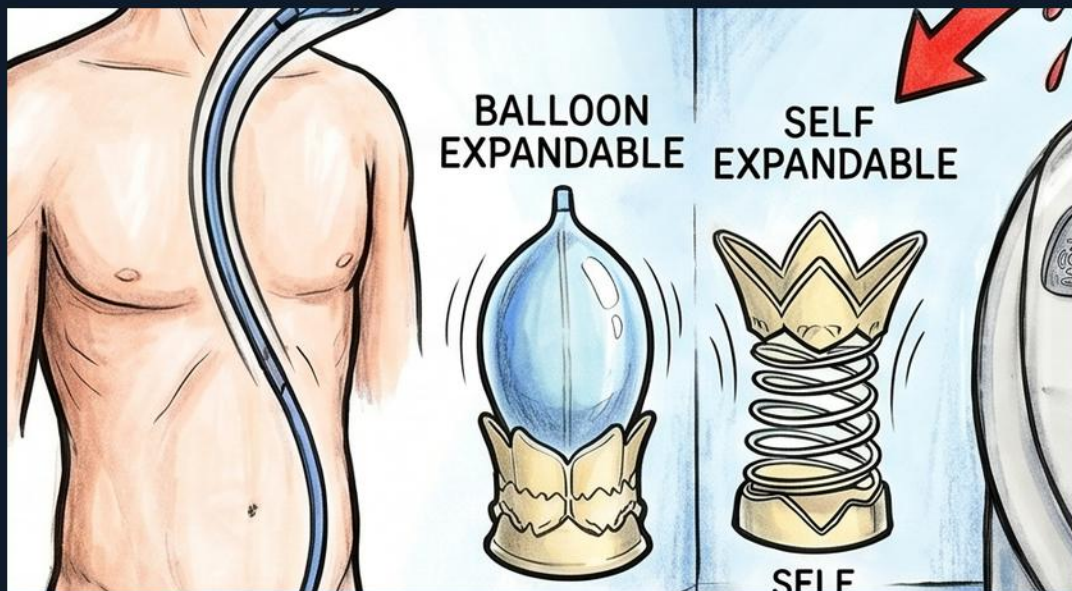
Right side labeled SAVR with a surgeon operating through an open chest.

WHAT IT ENCODES

Surgical aortic valve replacement is open-heart surgery via sternotomy on cardiopulmonary bypass, excising the native valve and sewing in a prosthesis. It allows direct repair and concomitant procedures but carries a bigger operative insult.

THE BOARD TRAP

Defaulting to SAVR for everyone — TAVI is now favored across many risk groups; SAVR is for specific scenarios.



Balloon-expandable valve

WHAT YOU SEE

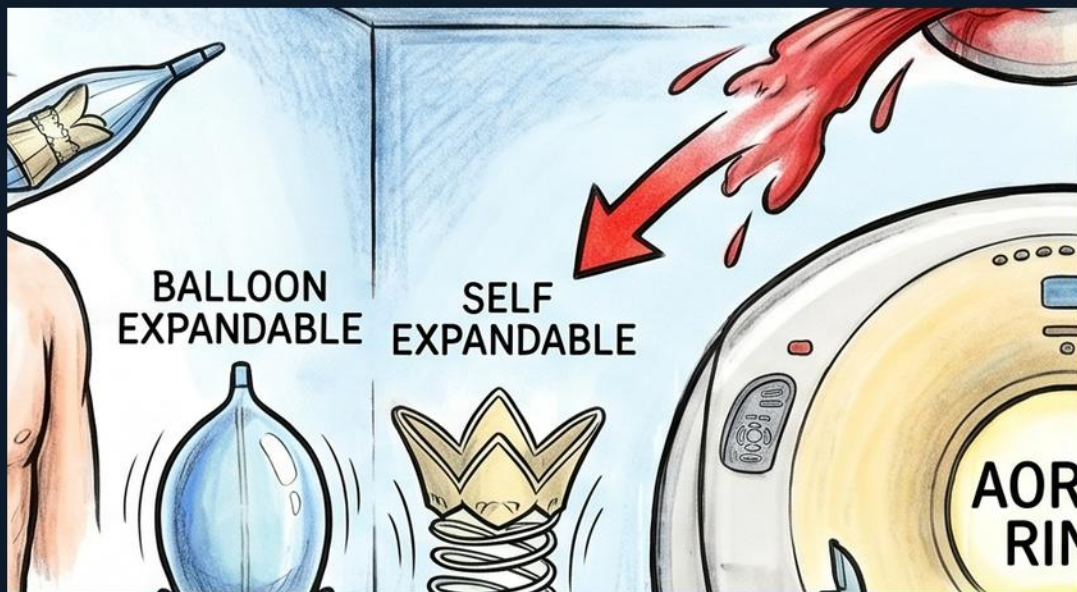
A balloon-mounted stented valve labeled BALLOON EXPANDABLE.

WHAT IT ENCODES

Balloon-expandable TAVI valves are positioned and then expanded by inflating a balloon, giving precise, controlled deployment. Valve type selection depends on annular anatomy and access.

THE BOARD TRAP

Treating all TAVI valves as identical — balloon-expandable vs self-expandable differ in deployment and conduction-block risk.



Self-expandable valve

WHAT YOU SEE

A crown-like nitinol frame labeled SELF EXPANDABLE.

WHAT IT ENCODES

Self-expandable valves use a nitinol frame that opens on its own as the sheath is withdrawn. They sit higher/deeper depending on design and are associated with a higher rate of conduction disturbance and pacemaker need.

THE BOARD TRAP

Ignoring valve type when predicting pacemaker risk — self-expandable valves carry the highest new AV-block rate.



CT for annulus sizing

WHAT YOU SEE

The CT scanner in the center with AORTIC RING / ANNULAR RING marked on the gantry.

WHAT IT ENCODES

Pre-TAVI gated cardiac CT measures the aortic annulus to size the prosthesis and plan access. Correct annular sizing minimizes paravalvular leak and prevents annular injury or valve embolization.

THE BOARD TRAP

Sizing TAVI off echo alone — CT annular measurement is the standard to avoid paravalvular leak and malsizing.



Paravalvular leak

WHAT YOU SEE

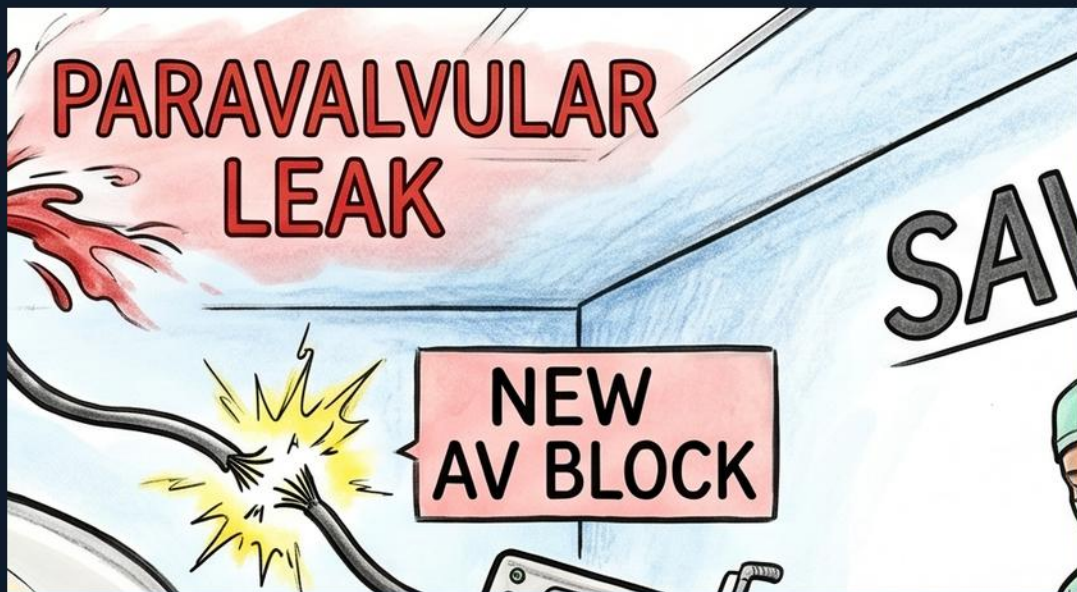
Top-center valve with a red arrow of blood squirting around its rim, labeled PARAVALVULAR LEAK.

WHAT IT ENCODES

Paravalvular (perivalvular) aortic regurgitation is leakage AROUND the prosthesis rather than through it, a hallmark complication of TAVI from incomplete sealing against a calcified, irregular annulus. Even moderate paravalvular AR worsens prognosis.

THE BOARD TRAP

Dismissing post-TAVI paravalvular leak — it is a key complication and more than mild regurgitation impairs survival.



New AV block

WHAT YOU SEE

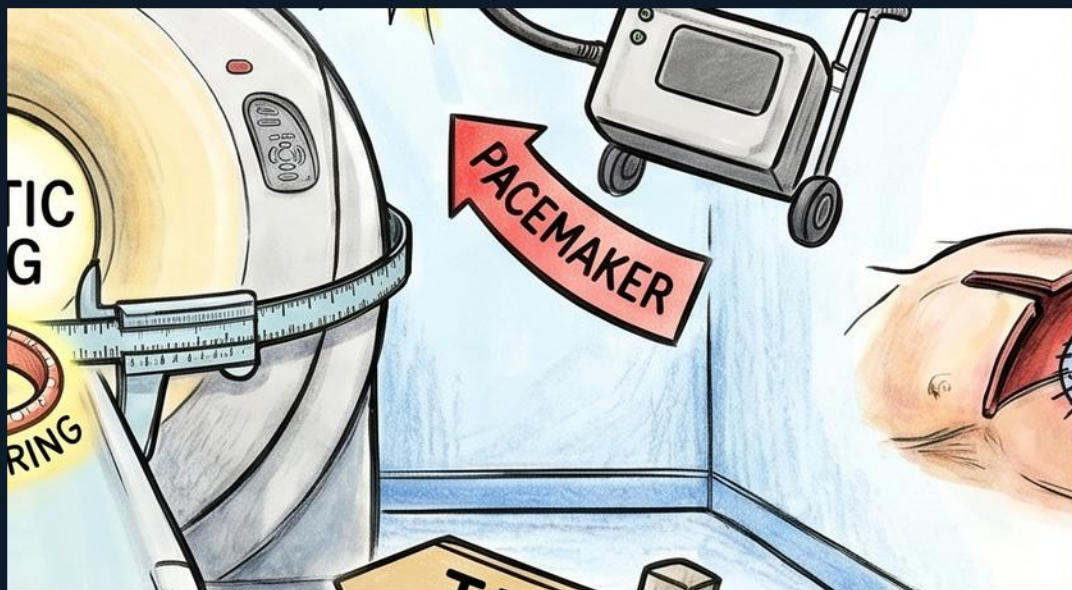
A lightning-bolt conduction symbol labeled NEW AV BLOCK upper right.

WHAT IT ENCODES

The valve sits directly over the conduction system, so TAVI can injure the AV node/His bundle and cause new high-grade AV block or left bundle branch block. This risk is higher with TAVI than SAVR and greatest with self-expandable valves.

THE BOARD TRAP

Underestimating conduction-block risk after TAVI — new high-grade AV block is common, especially with self-expandable valves.



Pacemaker requirement

WHAT YOU SEE

A cardiac monitor/pacer device labeled PACEMAKER center-right.

WHAT IT ENCODES

New high-grade AV block after valve replacement frequently requires permanent pacemaker implantation. Pacemaker rates are higher after TAVI than SAVR and must be discussed in patient consent and valve choice.

THE BOARD TRAP

Quoting equal pacemaker rates for TAVI and SAVR — TAVI (especially self-expandable) needs pacemakers more often.



Symptomatic severe AS, EF<50% → replace

WHAT YOU SEE

Podium with a gold star reading SYMPTOMATIC, EF UNDER 50, REPLACE.

WHAT IT ENCODES

Symptomatic severe aortic stenosis is a Class I indication for valve replacement, as is severe AS with LV systolic dysfunction (EF below 50%). Once these triggers are met, intervene — medical therapy does not change the natural history.

THE BOARD TRAP

Delaying intervention in symptomatic severe AS or AS with EF<50% — both are Class I indications to replace the valve.



TAVI favored

WHAT YOU SEE

A signpost labeled TAVI FAVORED in the center-right.

WHAT IT ENCODES

TAVI has expanded from high-risk to intermediate- and selected low-risk patients and is increasingly the default for transfemoral-suitable anatomy, especially older patients. The Heart Team weighs anatomy, age, and risk.

THE BOARD TRAP

Restricting TAVI to inoperable patients only — it is now favored across many risk groups, not just prohibitive risk.



SAVR: low-risk young

WHAT YOU SEE

Signpost arm reading LOW RISK YOUNG on the right.

WHAT IT ENCODES

Younger, low-surgical-risk patients are often steered to SAVR because of greater long-term durability data and concerns about lifetime management/valve durability with TAVI. Life expectancy and future re-intervention strategy drive this choice.

THE BOARD TRAP

Choosing TAVI reflexively for a young low-risk patient — SAVR durability often favors surgery in that group.



SAVR: bicuspid valve

WHAT YOU SEE

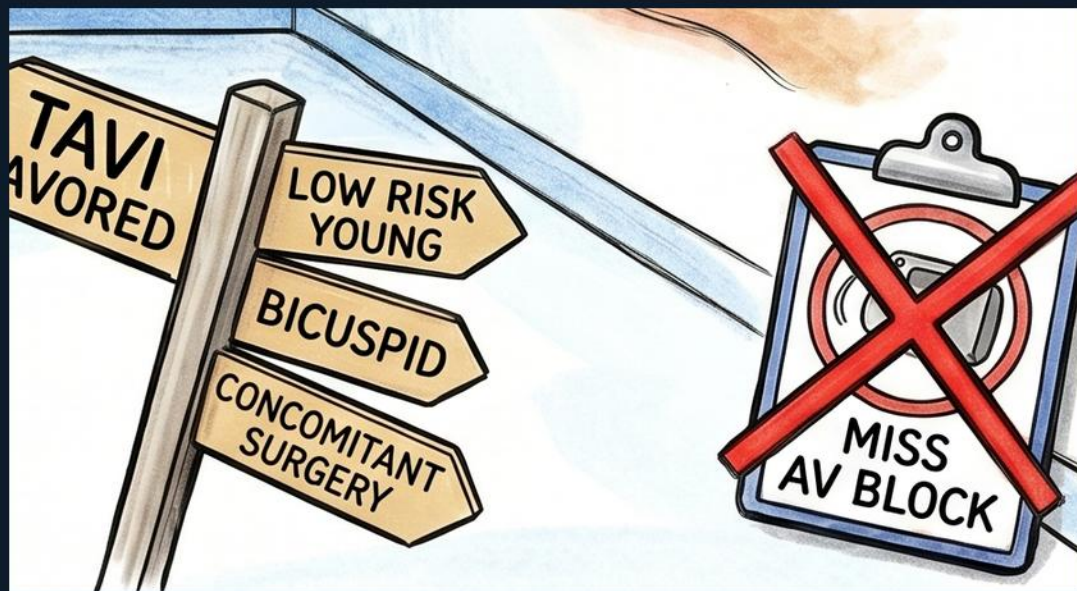
Signpost arm reading BICUSPID on the right.

WHAT IT ENCODES

Bicuspid aortic valve anatomy (asymmetric calcification, elliptical annulus, associated aortopathy) is less ideal for TAVI sealing and may favor SAVR, though TAVI is used in selected cases. Anatomy is a major decider.

THE BOARD TRAP

Ignoring bicuspid anatomy in valve choice — it raises paravalvular-leak risk and often tips toward SAVR.



SAVR: concomitant cardiac surgery

WHAT YOU SEE

Signpost arm reading CONCOMITANT SURGERY on the right.

WHAT IT ENCODES

When the patient already needs another cardiac operation (e.g., CABG or another valve), SAVR is performed together with that surgery rather than doing a separate TAVI. One operation addresses everything.

THE BOARD TRAP

Planning TAVI when the patient needs concomitant cardiac surgery — combine valve replacement with the surgery (SAVR).



Don't ignore the leak

WHAT YOU SEE

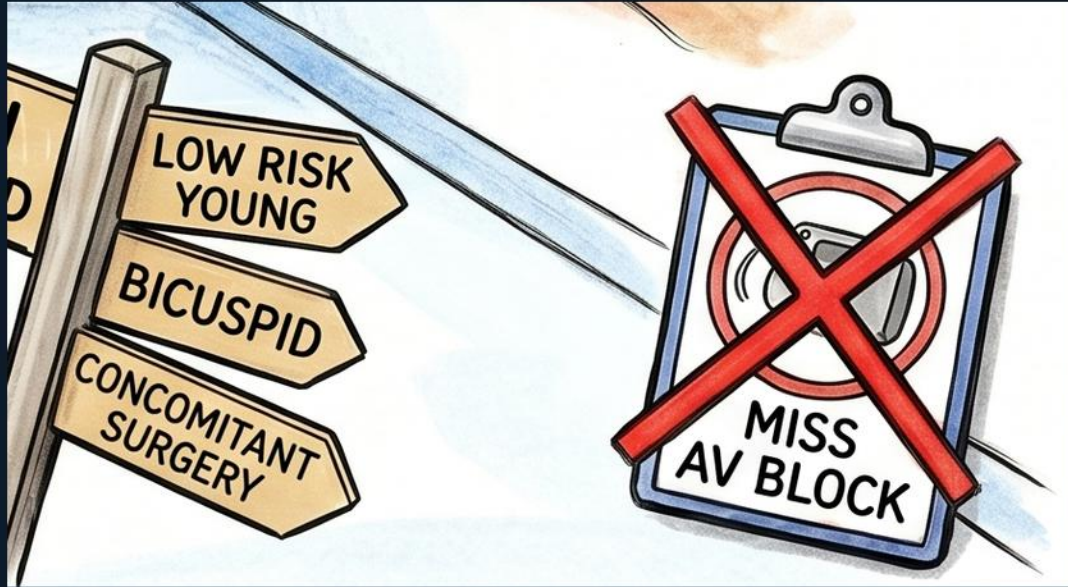
Bottom-left framed picture with a red X, captioned IGNORE LEAK.

WHAT IT ENCODES

A core teaching point: never dismiss post-procedure paravalvular regurgitation. More-than-mild paravalvular leak after TAVI is associated with worse outcomes and must be assessed and managed.

THE BOARD TRAP

Ignoring paravalvular leak after TAVI — it is a prognostically important complication, not a benign finding.



Don't miss AV block

WHAT YOU SEE

Bottom-right framed picture with a red X, captioned MISS AV BLOCK.

WHAT IT ENCODES

The paired warning: do not overlook new conduction disease after valve replacement. Monitor for high-grade AV block post-TAVI, as many patients need a permanent pacemaker.

THE BOARD TRAP

Missing new AV block after TAVI — vigilance for conduction disturbance and pacemaker need is essential.

Summary — every symbol

1. TAVI — transcatheter valve

Transcatheter aortic valve implantation delivers a prosthetic valve via a catheter (usually transfemoral), with no sternotomy and no cardiopulmonary bypass

3. Balloon-expandable valve

Balloon-expandable TAVI valves are positioned and then expanded by inflating a balloon, giving precise, controlled deployment

5. CT for annulus sizing

Pre-TAVI gated cardiac CT measures the aortic annulus to size the prosthesis and plan access

7. New AV block

The valve sits directly over the conduction system, so TAVI can injure the AV node/His bundle and cause new high-grade AV block or left bundle branch block

9. Symptomatic severe AS, EF<50% → replace

Symptomatic severe aortic stenosis is a Class I indication for valve replacement, as is severe AS with LV systolic dysfunction (EF below 50%)

11. SAVR: low-risk young

Younger, low-surgical-risk patients are often steered to SAVR because of greater long-term durability data and concerns about lifetime management/valve durability with TAVI

13. SAVR: concomitant cardiac surgery

When the patient already needs another cardiac operation (e.g., CABG or another valve), SAVR is performed together with that surgery rather than doing a separate TAVI

15. Don't miss AV block

The paired warning: do not overlook new conduction disease after valve replacement

2. SAVR — surgical replacement

Surgical aortic valve replacement is open-heart surgery via sternotomy on cardiopulmonary bypass, excising the native valve and sewing in a prosthesis

4. Self-expandable valve

Self-expandable valves use a nitinol frame that opens on its own as the sheath is withdrawn

6. Paravalvular leak

Paravalvular (perivalvular) aortic regurgitation is leakage AROUND the prosthesis rather than through it, a hallmark complication of TAVI from incomplete sealing against a calcified, irregular annulus

8. Pacemaker requirement

New high-grade AV block after valve replacement frequently requires permanent pacemaker implantation

10. TAVI favored

TAVI has expanded from high-risk to intermediate- and selected low-risk patients and is increasingly the default for transfemoral-suitable anatomy, especially older patients

12. SAVR: bicuspid valve

Bicuspid aortic valve anatomy (asymmetric calcification, elliptical annulus, associated aortopathy) is less ideal for TAVI sealing and may favor SAVR, though TAVI is used in selected cases

14. Don't ignore the leak

A core teaching point: never dismiss post-procedure paravalvular regurgitation